BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 155 · Lesson 155 · PageRank: Iterating a Link Graph

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does the inverted index map?

Q2. What is the role of a router?

Q3. What does a cache hit mean?

Q4. When can binary search be used?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

PageRank — Link-based iterative importance algorithm.

directed graph — Nodes joined by one-way edges.

outdegree — Number of a page's outgoing hyperlinks.

damping — Factor balancing link following and random jumping.

New learning

Build graph — Web pages are vertices; directed hyperlinks are edges.

Distribute rank — Each page shares rank across its outgoing links.

Apply damping — PRnew=(1-d)/N+d×sum(incoming PR/outdegree).

Worked example

A→B,C; B→C; C→A. Initially PR(A)=PR(B)=PR(C)=1/3; d=0.85. Calculate one iteration.

1. A→B,C | B→C | C→A — Identify the input or diagram and predict the first result.

2. Incoming shares 1/3,1/6,1/2 — Follow the second stage and explain its importance.

3. Result .333,.192,.475 — Incoming shares A:1/3; B:1/6; C:1/6+1/3=1/2. Base (1-.85)/3=.05. New A=.3333, B=.1917, C=.4750, sum=1.0000.

We Do · Practise with guidance

A→B; B→A,C; C→A; equal initial rank and d=0.85. Which page ranks highest after one update?

A→B | B→A,C | C→A

Identify the input or diagram and predict the first result.

Shares 1/2,1/3,1/6

Follow the second stage and explain its importance.

A=.475 highest

Incoming contributions A=1/6+1/3=1/2, B=1/3, C=1/6. Add base 0.05 and scale by 0.85: A=.4750, B=.3333, C=.1917. A ranks highest.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Explain how PageRank uses inbound links and outbound link counts. [6]

Question 2

A page of rank 0.40 has two outgoing links. What undamped contribution goes down each? [4]

Question 3

Compute one PageRank update for A→B,C B→C C→A with equal initial ranks and d=.85, then explain an important limitation. [8]

Exit Ticket · Answer before checking

With PageRank a page with two outgoing links shares its current rank equally between them in the simplified model.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Search terms to document IDs.

Q2. Forward packets between IP networks.

Q3. Required data already present in cache.

Q4. On sorted data.

You Do pages · Independent questions

Q1. Incoming links carry rank [2], each source shares across outgoing links [2], iterative importance measures [2].

Q2. 0.40/2=0.20 to each [4].

Q3. Incoming shares [3], updated scores A=.3333, B=.1917, C=.475 [3], ignores query-text relevance/content [2].

Exit Ticket · Model answer

True.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: